Exercise for MA-INF 2213 Computer Vision SS25
09.05.2025
Submission deadline: 08.06.2025

Important: We recommend using **Python 3.9** at least for implementing your solutions in this exercise. We recommend to use **PyTorch (version 2.3)** and **torchvision (version 0.18)**.

Grading: Submissions that generate runtime errors or produce obviously rubbish results (e.g., nans, inf, or meaningless visual output in future exercises) will receive at most 50% of the points.

Plagiarism: Plagiarism in any form is prohibited. If your solution contains code copied from any source (e.g. other students, or solutions from the web), you will receive **0 points** for the entire exercise sheet.

Submission: You can complete the exercise in groups of two, but only one submission per group is allowed. Include a *README.txt* file with your group members in each solution. Points for solutions without a readme file will only be given to the uploader.

Image classification on CIFAR-10 dataset

The objective of this assignment is to identify optimum hyperparameters for a given set of basic operators covered in the lecture and learn to arrange them to improve the provided LeNet sequential architecture. You are allowed to use different layers covered in the lecture such as convolution, activations, pooling, normalization, fully connected, residual blocks, etc. However, usage of any kind of existing layers which are not covered in the lecture is strictly prohibited. But you are free to combine the elementary operators covered in the lecture in whatever way you want. But the motivation behind such an architectural choice should be clearly covered in the report. For example, using a certain scaling, without explaining the reason, may cause a mark deduction (assume that the reviewer does not know anything outside the course material).

Regarding the code

We have provided a notebook for all tasks. The notebook already includes the datasets, dataloaders, train, and test loops. To ensure reproducibility, we fix the seed for the random number generators (please don't change this part).

As for the Task 1-4, The task is to complete the required cells with your code to implement several improvements to the base architecture. Please carefully read the instructions for each task.

As for the Task 5, you should modify the base architecture to get the best possible accuracy. You can also add your own augmentations, modify the learning rate, batchsize, and change the optimizer parameters to your liking.

Submission

Technical report (maximum of 3 pages):

The technical report is expected to briefly describe the details of the implementation and a detailed discussion of the results obtained (e.g., the effect of different modifications and your observations after applying them). A latex report template is included. It is recommended that a technical report include the following information (these points are not the suggested sections of a report, but the list of information to be covered):

- **Introduction:** Brief description of the assignment problem, the architecture used, and the methods you are going to experiment with.
- **Experiments:** A major part of the report can be dedicated to experiments and discussions. Experimental results (accuracy, training time, number of parameters, etc.) related to each question (Tasks 1-5) should be presented separately. To effectively represent the results, you can use tables, graphs, figures, etc.
- **Discussion (can be merged with experiments):** The report is expected to have a thorough analysis and a detailed discussion about the results obtained for each question.
- **Conclusion:** Brief description of all the work done, and mentioning important observations and remarks.

Completed Code:

There is a skeleton code provided as a part of the assignment. Submit the completed cells for task 1-4. As for task 5, you can organize your code according to the following suggested file structure:

- **model.py:** Contains the implementation of your neural network.
- **train.py:** Contains the codes for training the model.
- **dataset.py:** Handles loading and preprocessing of the data.
- **eval.py:** Script to build the model, initialize it with pre-trained weights, and conduct evaluation.

Alternatively, you can submit the same completed notebook with the addition of **model.py** and **eval.py** as separate cells.

Notes:

- The maximum allowed number of parameters for your model is **120k**.
- The maximum number of epochs is **30**.
- You are allowed to report the best accuracy across all 30 epochs by mentioning the exact epoch number.

- Try to reach a minimum accuracy of 80% on the test set.
- If you are using additional libraries, include a requirements.txt file.
- You are not allowed to use pre-existing model implementations from PyTorch/torchvision or any library that includes implementations of classification models.
- Do not include the data in your submission.
- If you are using colab, write "!pip install torch==1.8.0"

Grading

You can earn 20 points from this exercise sheet.

Task 1: Modifying maxpooling layer

Modify the following network by changing every maxpool layer with *avgpool* layer. Modify the following network by removing maxpool layers and utlize convolutions with stride > 1. *(2 points)*

Task 2: Adding Dropout to the Base Network

Please modify the following network by introducing the dropout layer(s). *(1 points)*

Task 3: Adding Batchnorm Layer(s) to the Base Network

Please modify the following architecture by introducing Batchnormalization layers. You may change batch size, learning rate, and/or momentum. *(2 points)*

Task 4: Applying Dropout and Batch normalization on the base network

Please modify the following network by adding **both** dropout and batch normalization layer/s. *(1 points)*

Task 5: Finalize your network architecture

Using the previous observations and what you learned in the lecture, modify the following architecture to achieve the best possible accuracy. *(7 points)*

Technical report (maximum of 3 pages)

The technical report is expected to briefly describe the details of the implementation and a detailed discussion of the results obtained. *(7 points)*

Bonus point: Three submissions that achieve the best classification accuracy in the test set will receive 2 bonus points.